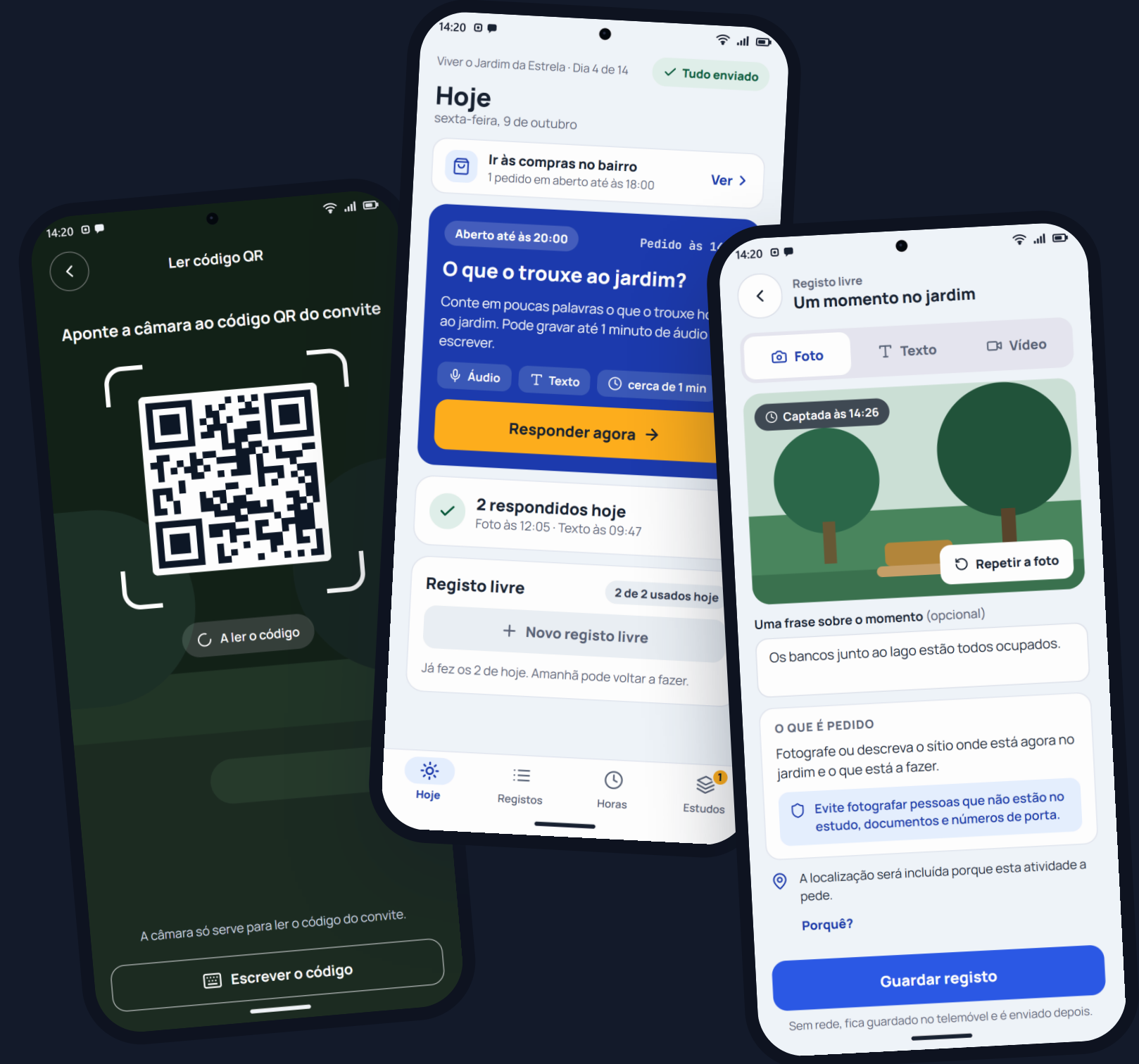




# Record life as it happens

A distributed, fault-tolerant platform for mobile ethnography and diary studies

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BSc in Computer Engineering · IADE, Universidade Europeia



## THE PROBLEM

# An entry made later is no longer the same

### Out in the street, offline

The entry has to stay saved on the phone.

### Everyone at once

A prompt sent to everyone brings a burst of photos and videos.

### PAPER DIARY

Participants said

90%

Actual adherence

11%

Stone et al., BMJ, 2002

WHO IT IS FOR

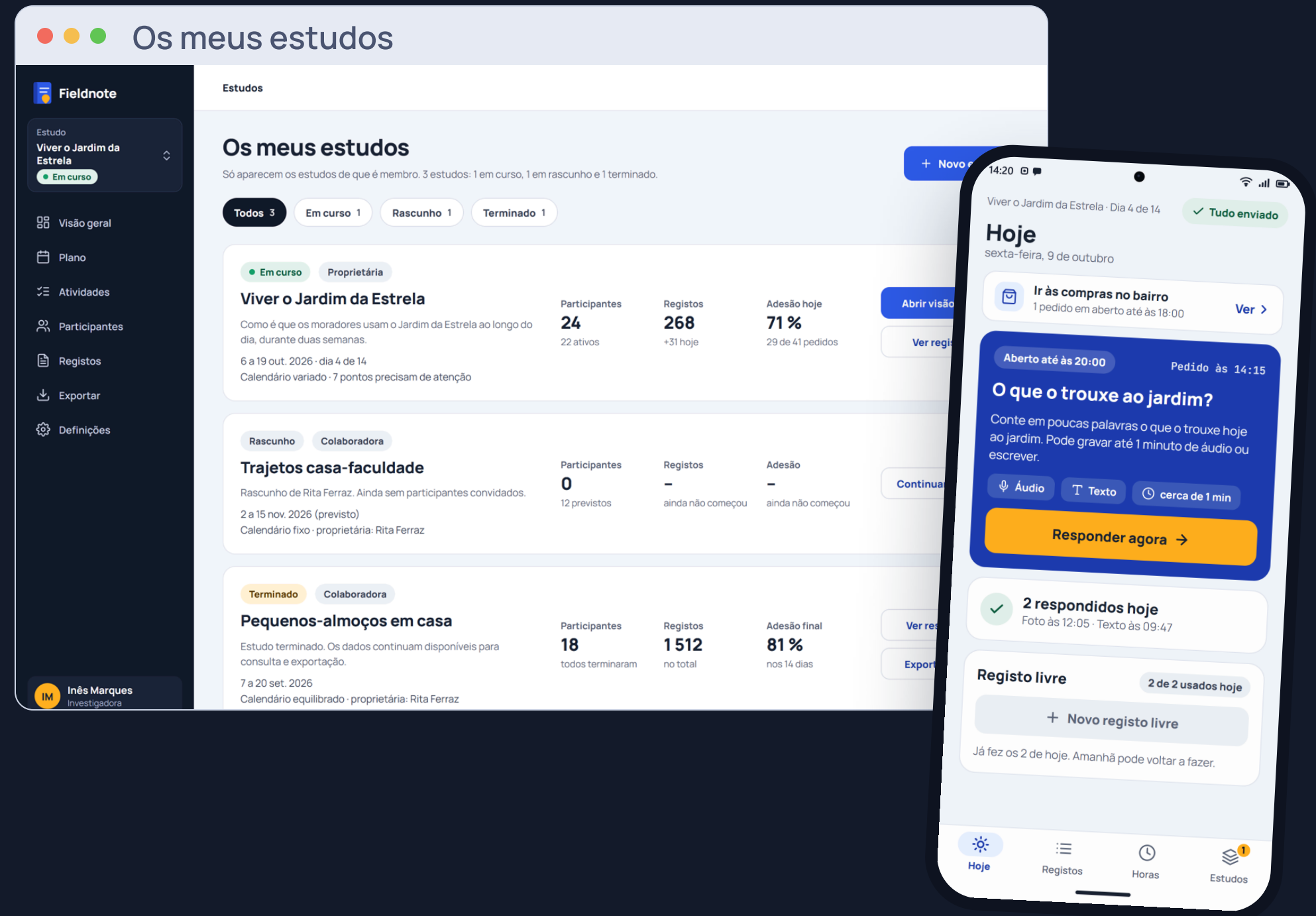
# Research in design and communication

The framing is UNIDCOM/IADE, the IADE research unit.

Data under the institution's control

Photo, video, audio, text and scale

App and consent in Portuguese



WHAT ALREADY EXISTS

Each feature exists elsewhere

| PLATFORM               | TYPE             | HOSTED BY                               | FOCUS                                       |
|------------------------|------------------|-----------------------------------------|---------------------------------------------|
| Indeemo, dscout, EthOS | Commercial       | The vendor                              | Rich media, recruitment                     |
| m-Path                 | Academic, closed | The vendor (Azure, Frankfurt)           | EMA questionnaires                          |
| Beiwe, AWARE           | Open source      | Whoever installs it (Beiwe only on AWS) | Phone sensors                               |
| Fieldnote              | Academic project | The institution, on its own server      | Rich media, fault tolerance, CSP scheduling |

What sets Fieldnote apart: the institution hosts the data.

THE SOLUTION

One system, three roles

ANDROID APP

Participant

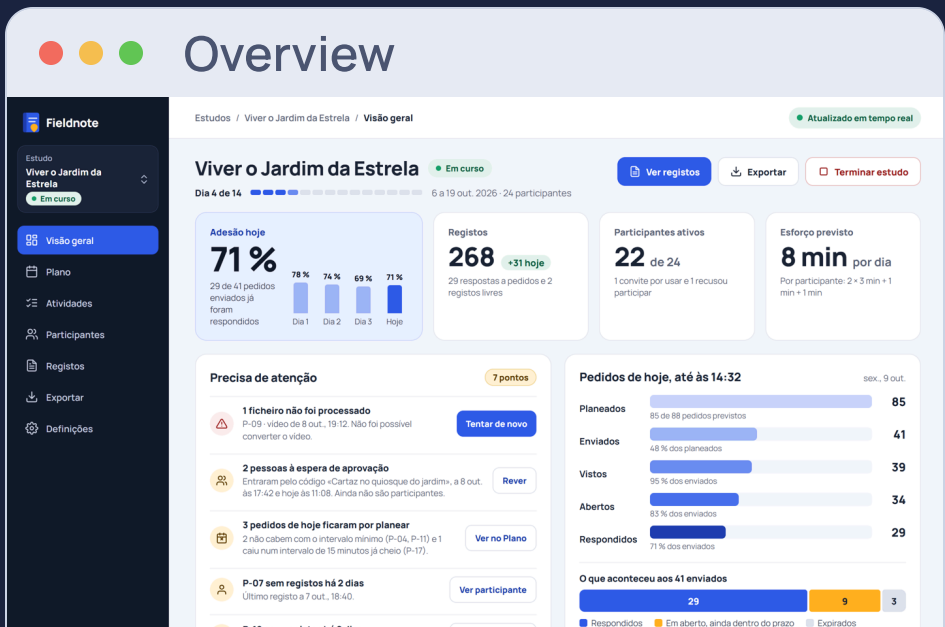
Joins a study, receives prompts and records entries.



WEB PORTAL

Researcher

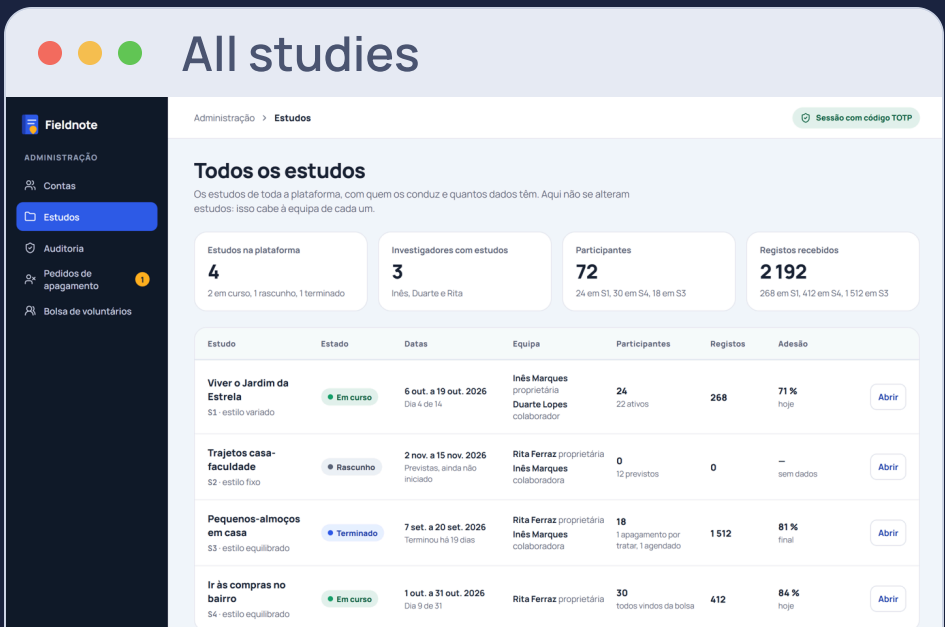
Creates the study and follows it live.



WEB PORTAL

Administrator

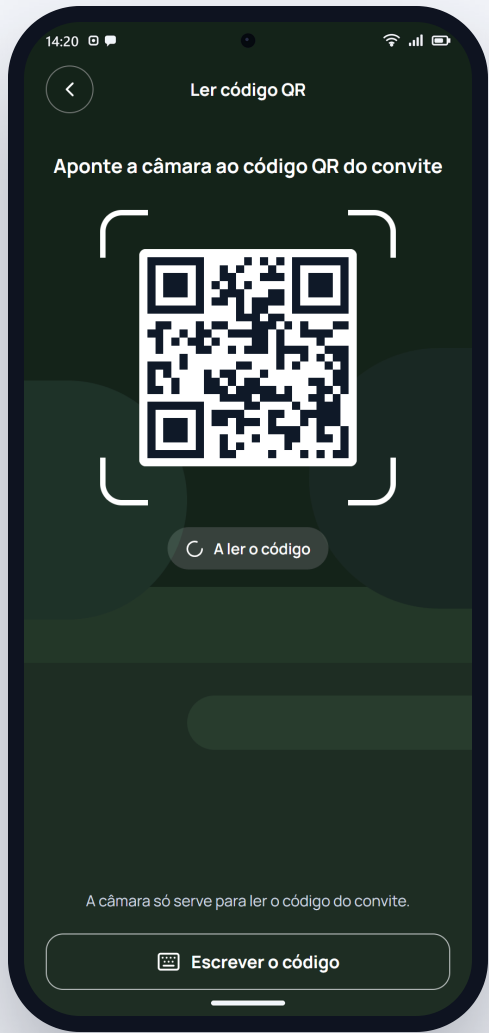
Manages accounts, studies and deletions.





PARTICIPANT

Three ways to join a study



Personal invite code  
Scanned as a QR



Or typed  
12 characters, single use



Volunteer pool  
Invitation in the app



Poster  
8 characters, reusable



## PARTICIPANT

# One prompt, one tap, one entry

The notification arrives, the app opens on Today and the participant chooses how to respond.

Text

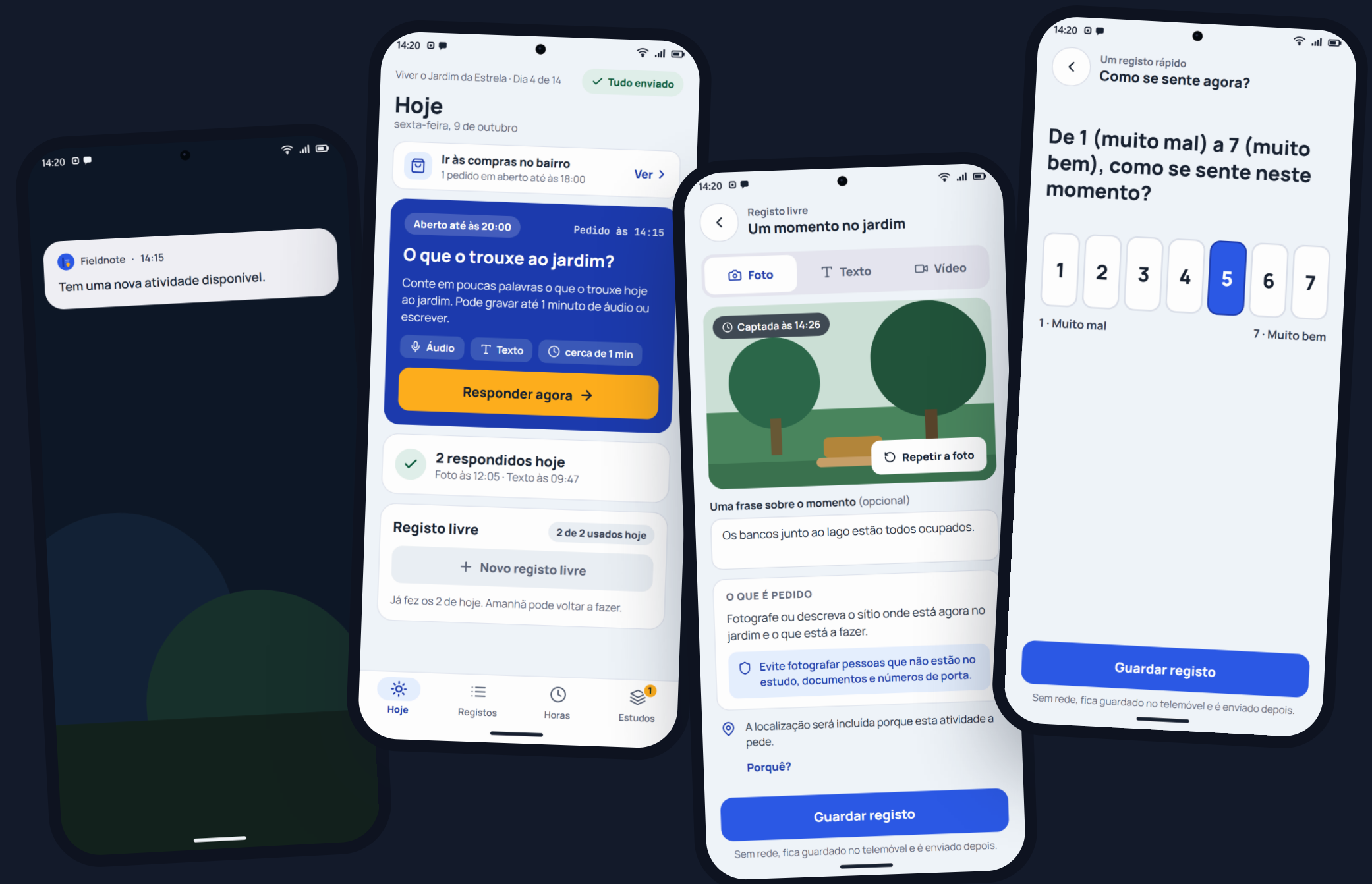
Photo

Video

Audio

Scale from 1 to 7

Multiple choice



## PARTICIPANT

# Without a network, the entry is kept

The phone keeps it in a local queue and resends when there is a connection. Resending does not duplicate.





PARTICIPANT

# The participant is in control

- 1

Consent before starting

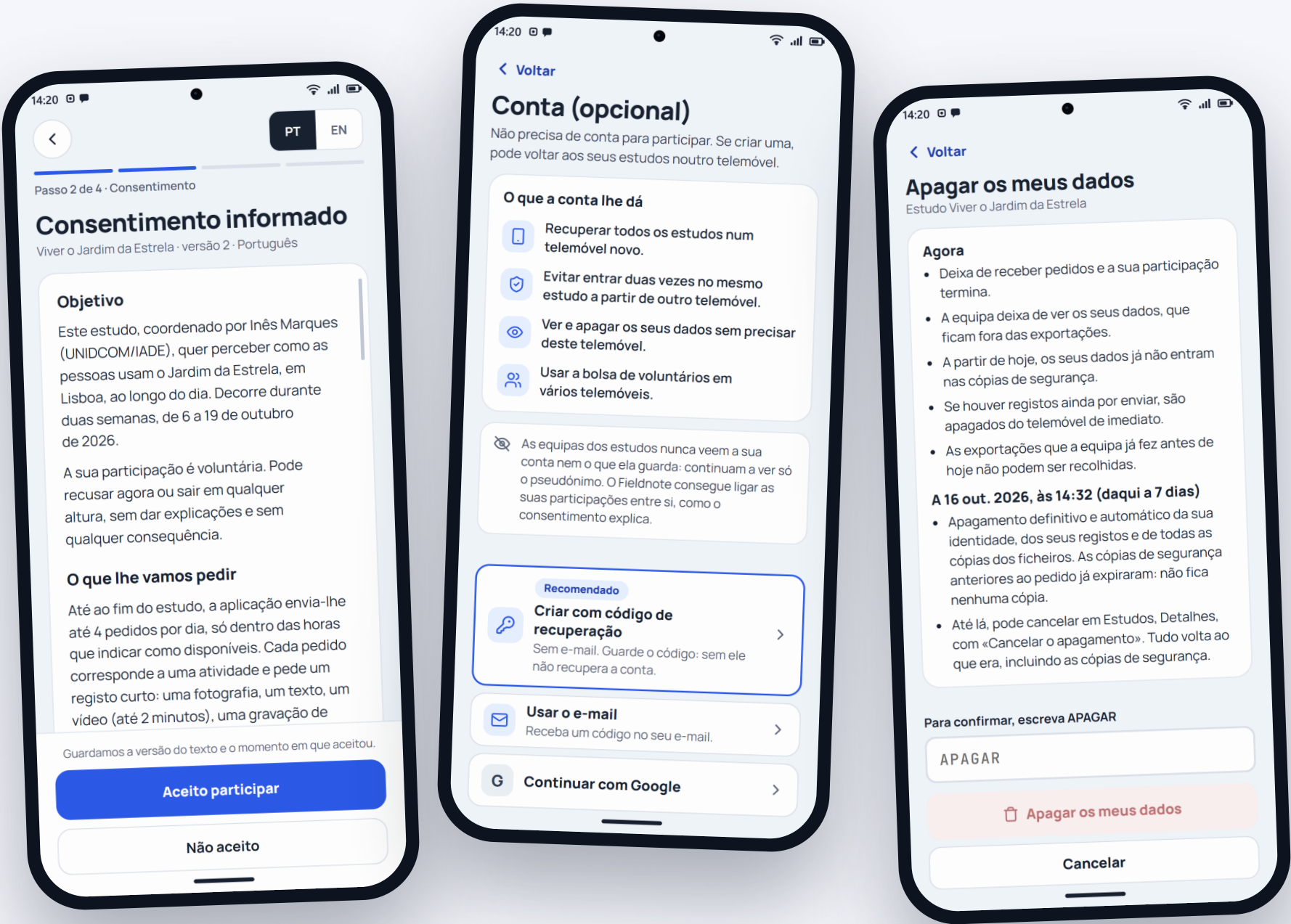
The full text is always available.
- 2

Optional account

Recovery code, email or Google, to switch phones.
- 3

Delete my data

Hidden at once, deleted on day 7.



RESEARCHER

# Create the study and its activities

## The study

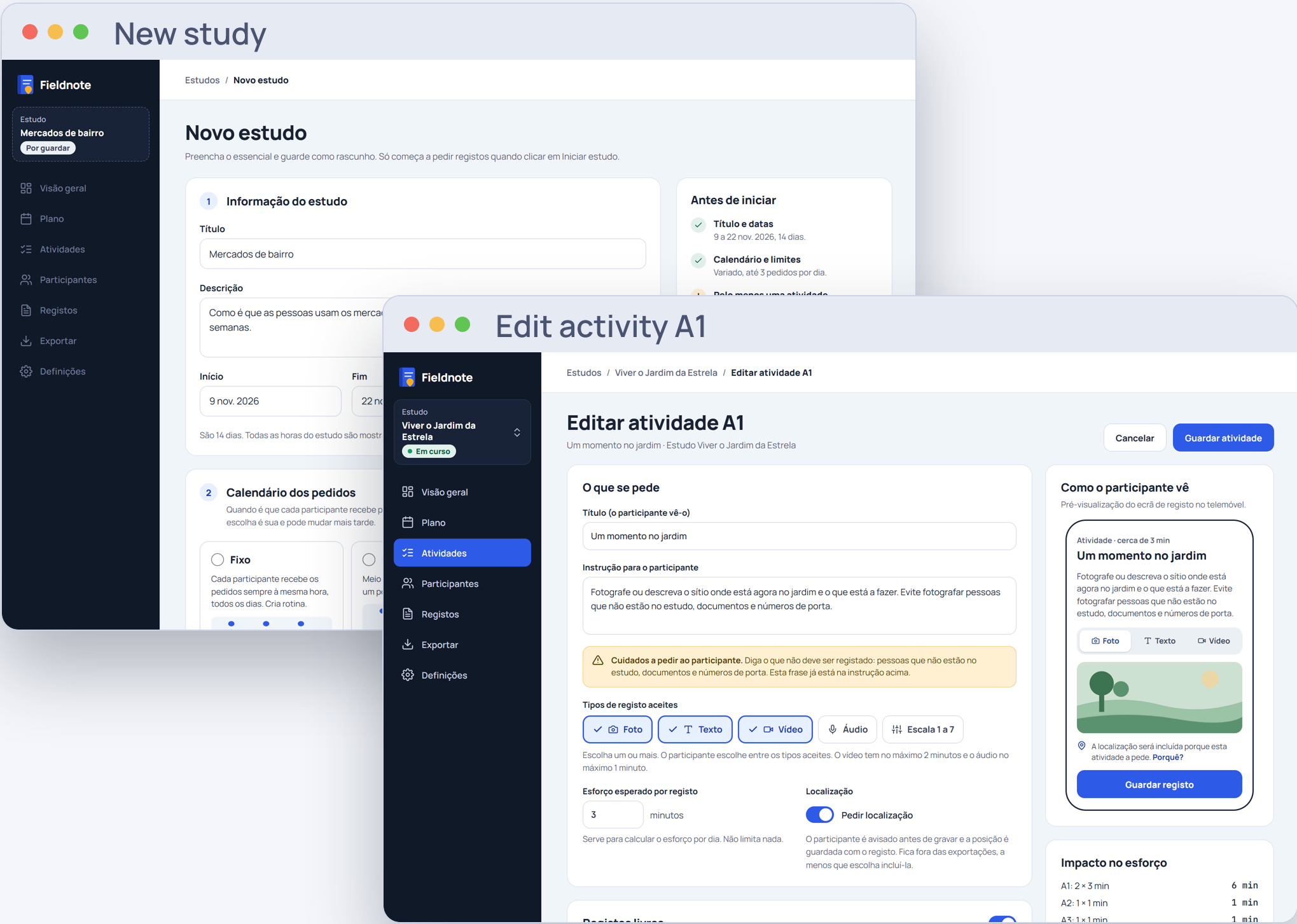
Title, dates and time zone.

## The activities

What to ask for, and in which format.

## The rules

When to ask, with daily limits.



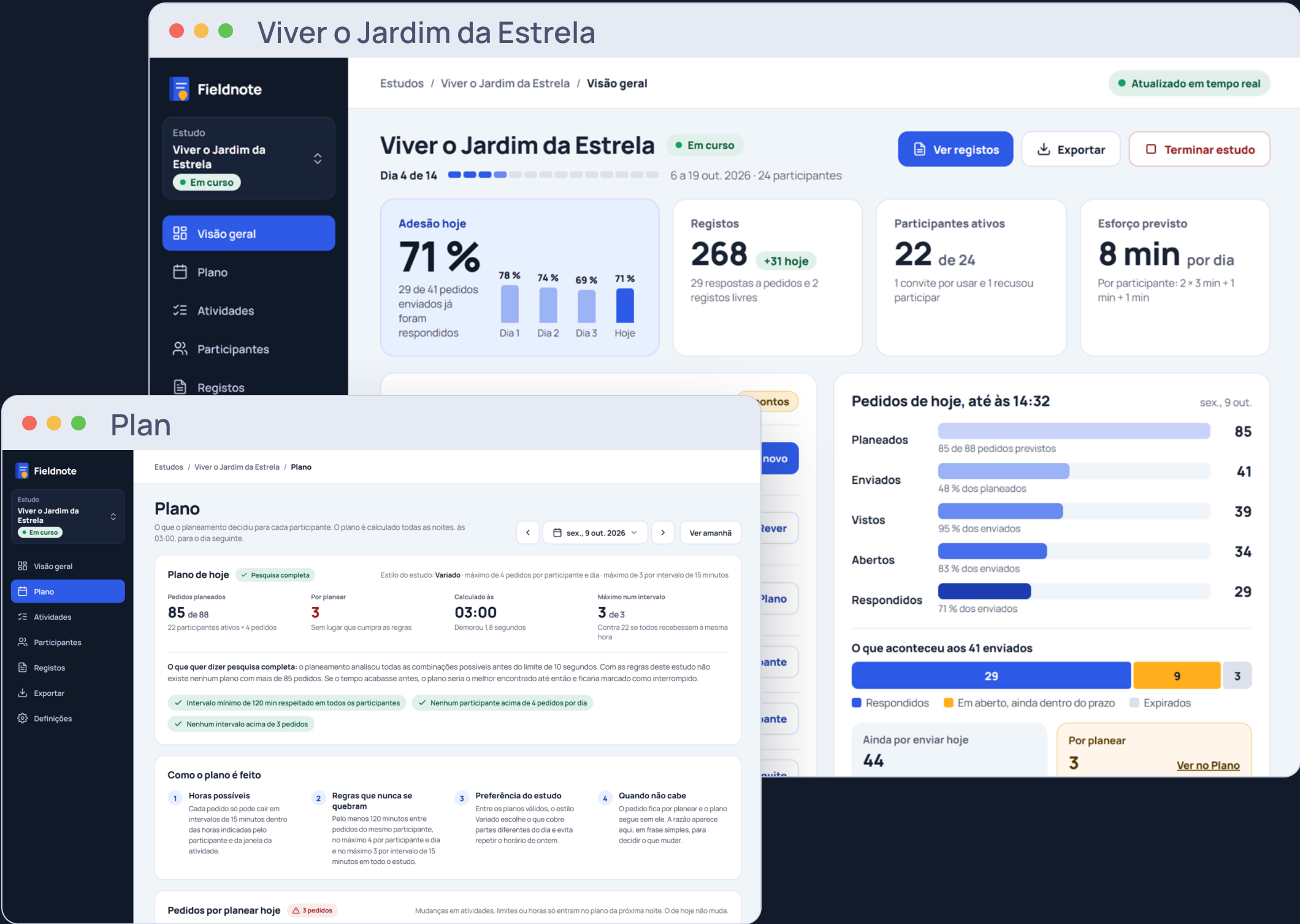
RESEARCHER

# Follow the study live

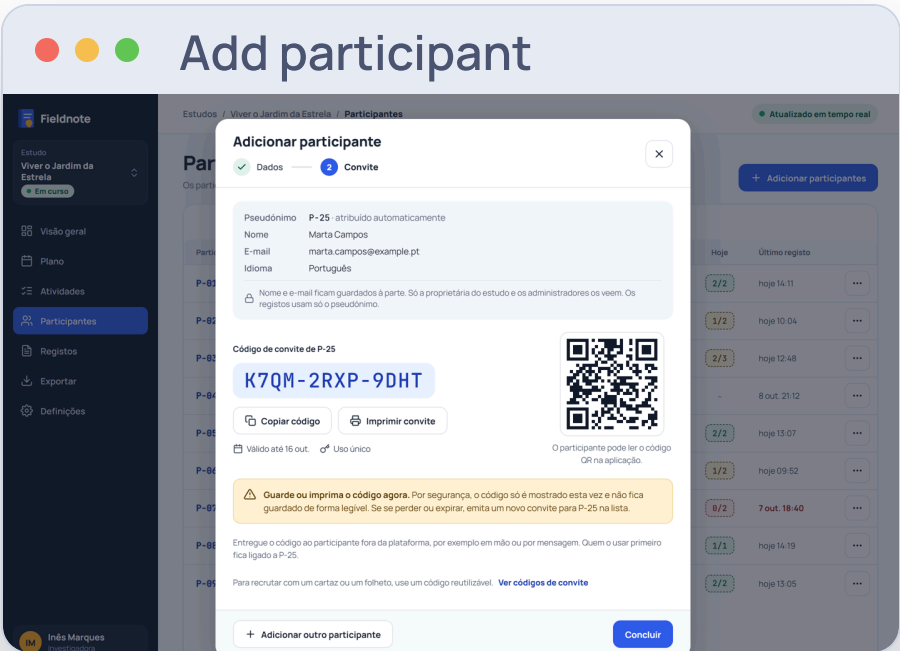
71% adherence today

268 entries received

22/24 active participants



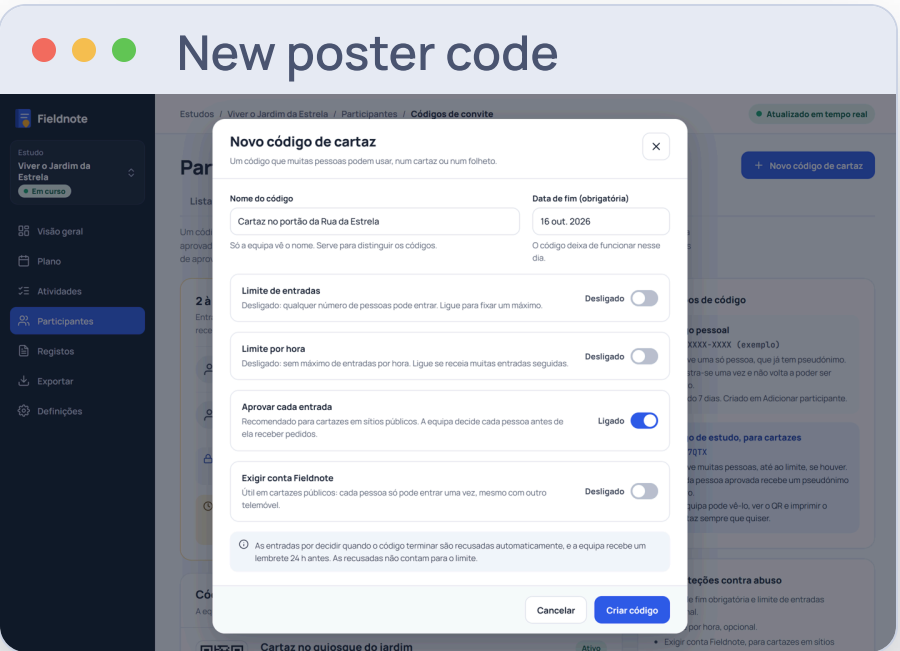
# Three ways to recruit



## Personal invite

A 12-character code for each person.

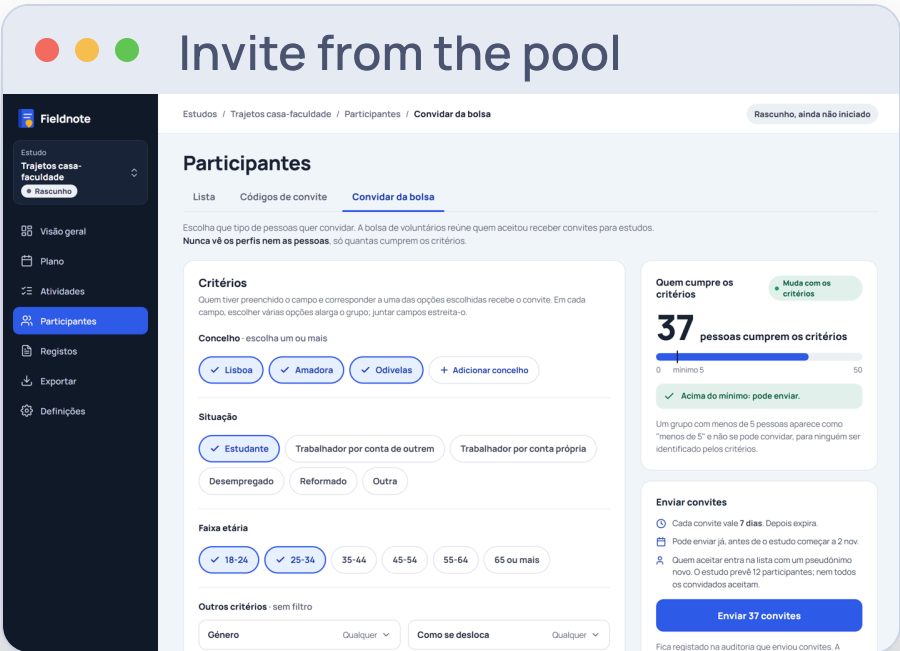
K7QM-2RXP-9DHT



## Poster in the street

Reusable study code, with an entry limit and an end date.

JE4K-7QTX



## Volunteer pool

The researcher only sees how many people meet the criteria.

37 people



ADMINISTRATOR

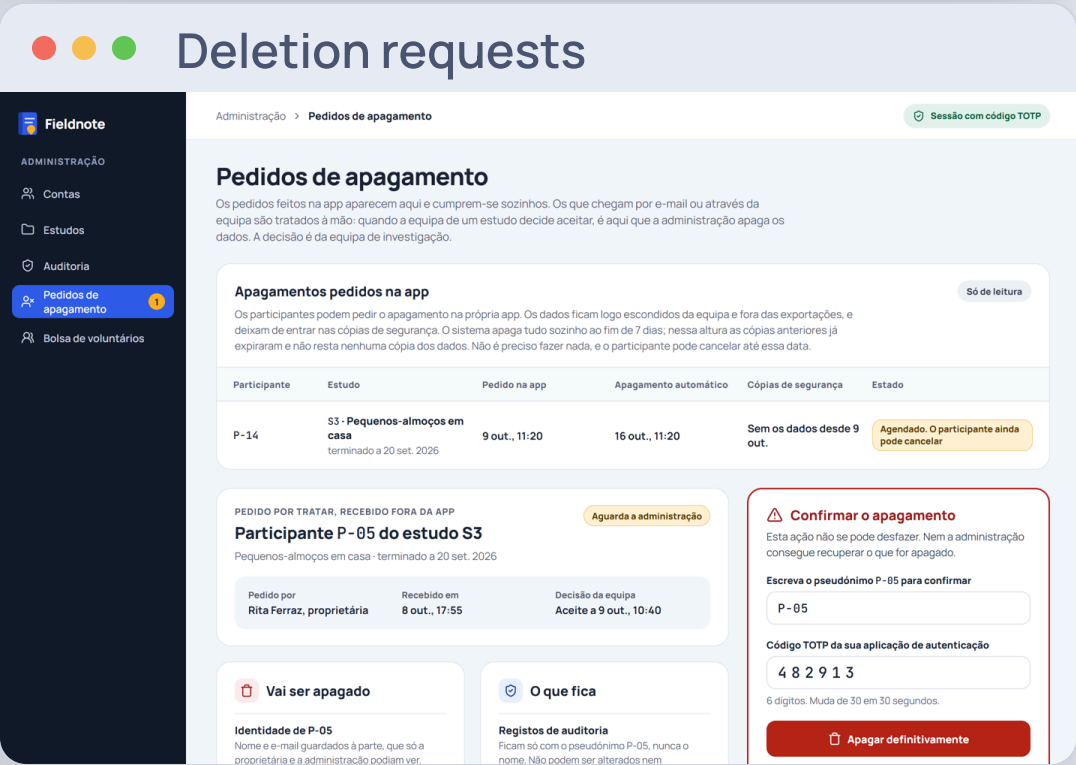
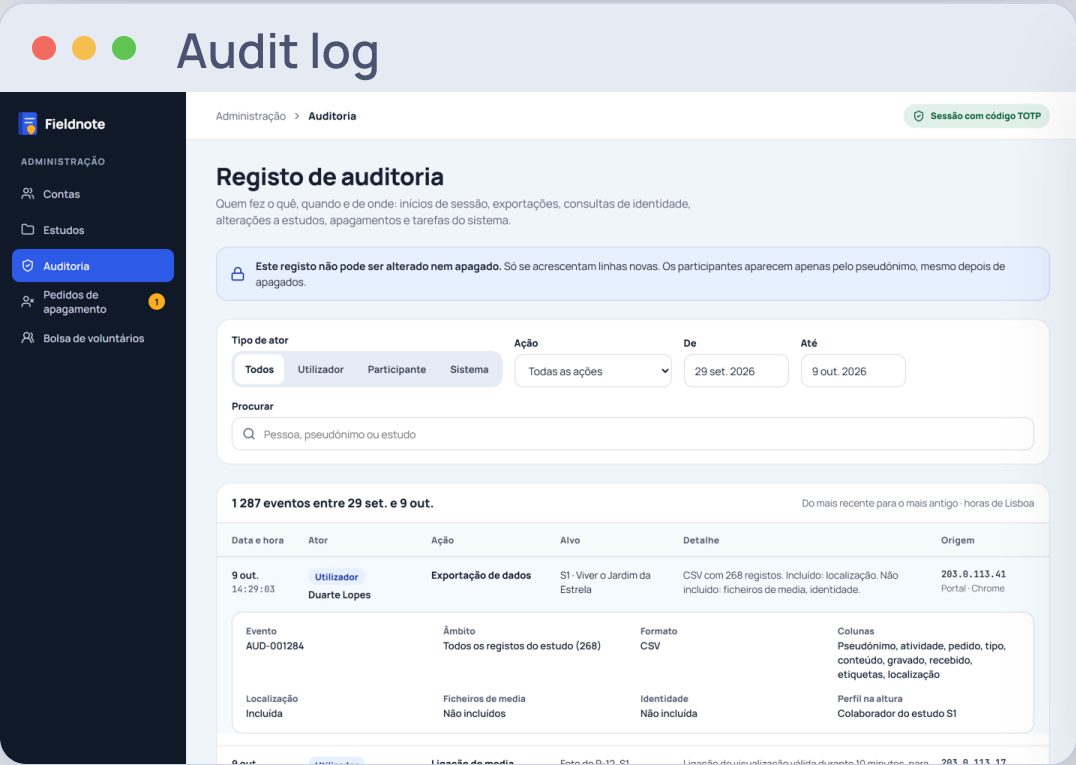
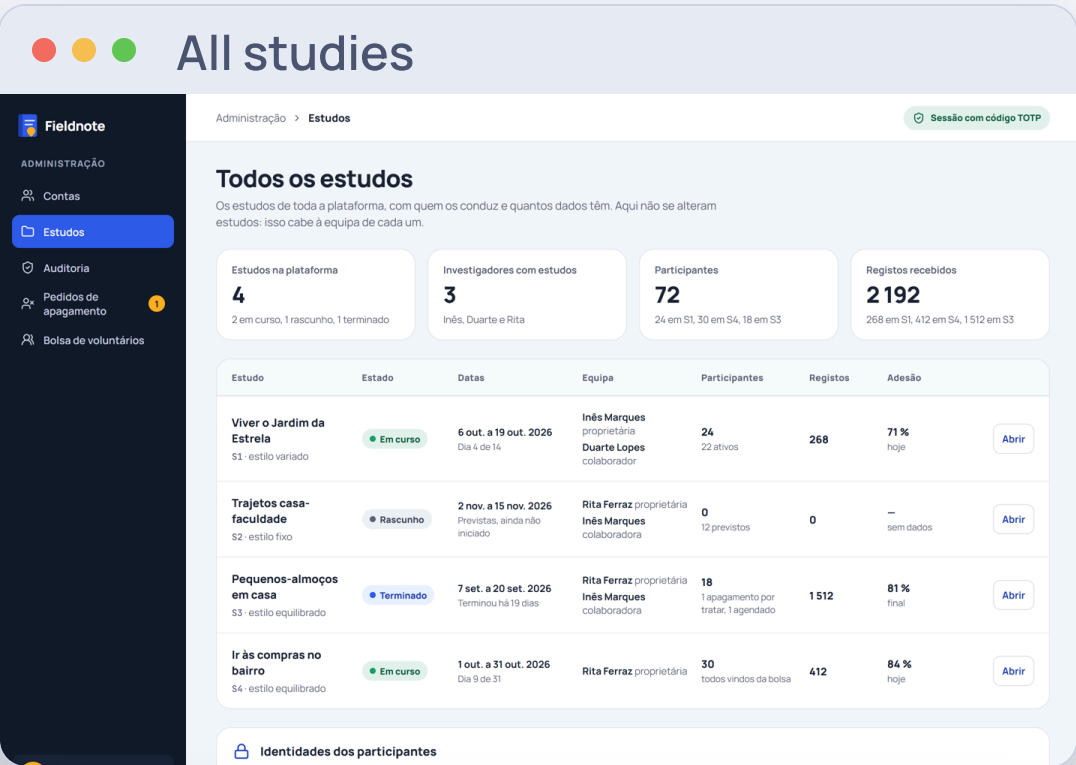
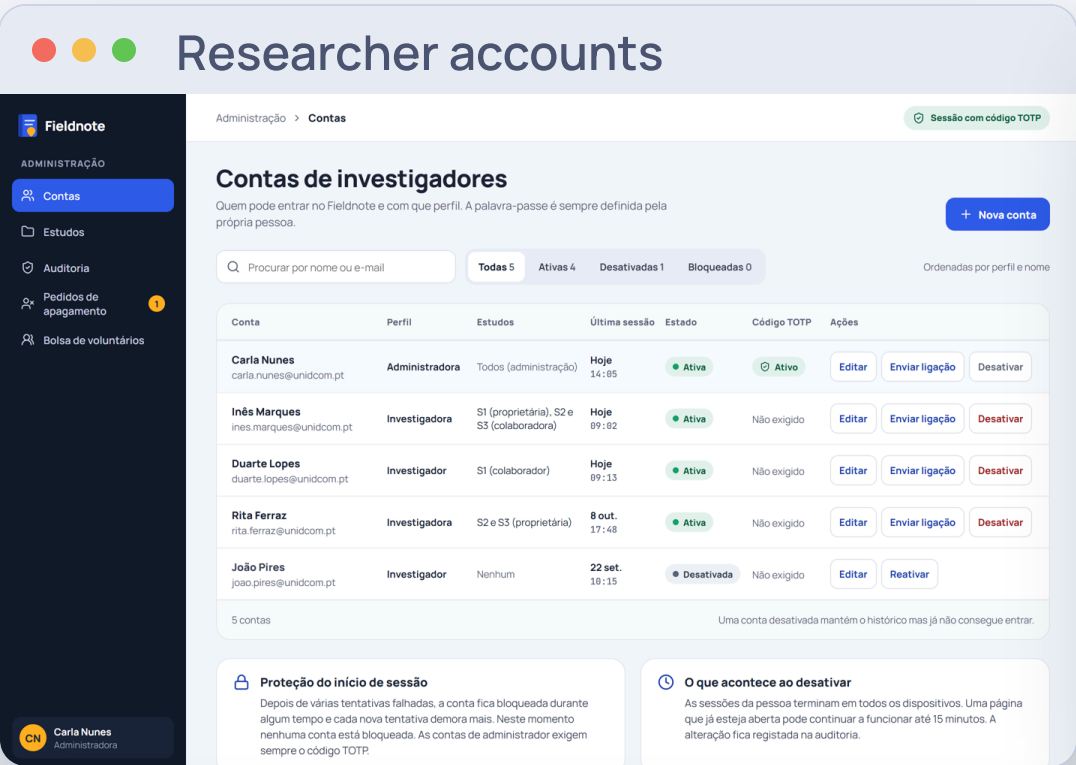
Oversee the platform

Researcher accounts, with sign-in by TOTP code

All studies on the platform

Volunteer pool

Audit log and deletion requests



## ARCHITECTURE

# Three nodes, one member of each cluster on each



One Docker host (Compose). Access only through the private network.



## FAULT TOLERANCE

# If a node goes down, the system keeps going



**Node 1**

Receiving entries



**Node 2**

Down



**Node 3**

Receiving entries

Keeps receiving entries if any node goes down.

A confirmed entry is never lost.

## FAULTS WE WILL INJECT IN THE DEMONSTRATION

Stop containers

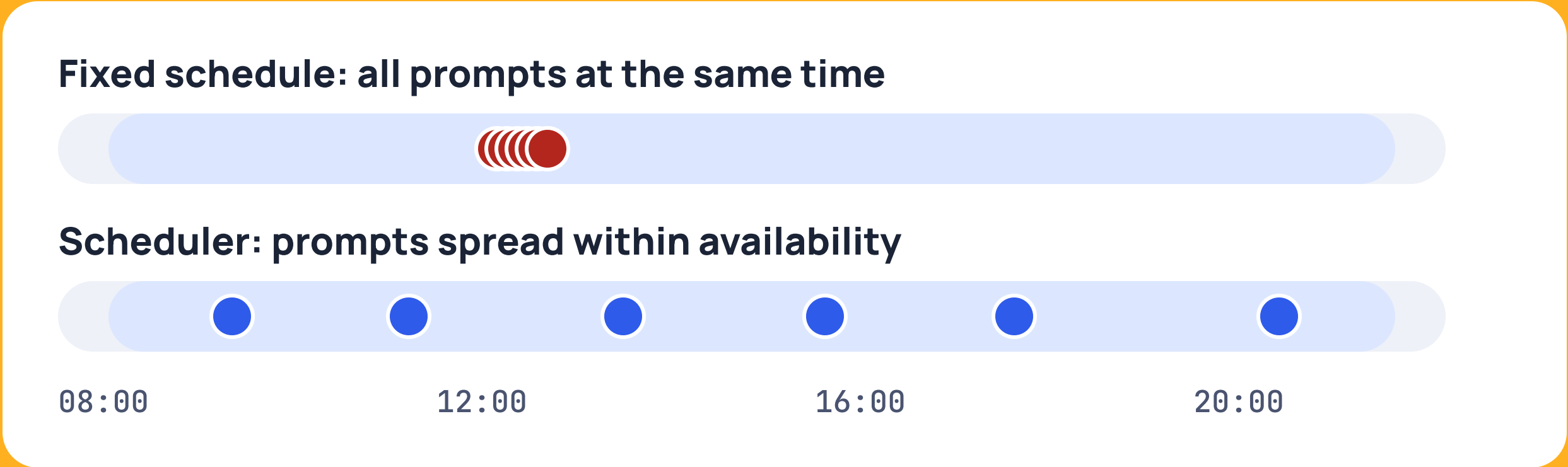
A whole node

The database primary

A storage node

Isolate a node from the network

# When to ask each person for each activity



- Availability
- Activity window
- Minimum gap
- Daily limit
- Max. at once per 15 min



Approved by the AI  
lecturer on 1 October  
2026

SECURITY

Protect the people who take part



Authentication and roles

The administrator signs in with a TOTP code.



Protected media

Signed, short-lived links.



Audit log

Sensitive actions are logged.



Pseudonym

Participants appear as P-07.



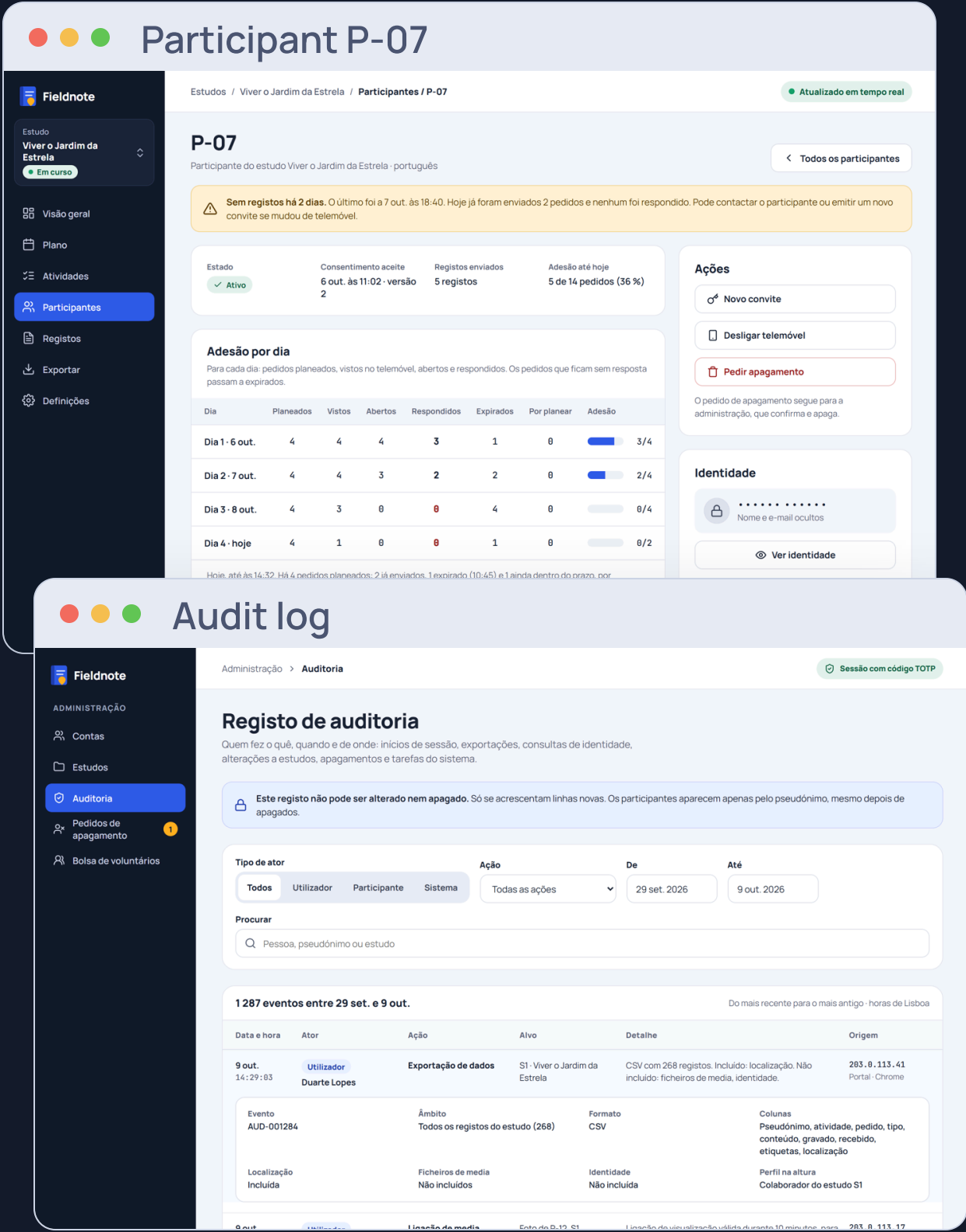
Photos without location

EXIF data is removed.



Data deletion

Fully deleted on day 7.



## WHAT THE RESEARCH CHANGED

# We read critically, and the design changed

### 01 Recorded time and received time

Paper diaries were filled in afterwards.

### 02 Daily limit chosen by the researcher

Two meta-analyses disagree on frequency.

### 03 Several entry formats per activity

Each person prefers one format, and recording audio in public is awkward.

### 04 Local backup notifications

Push is delivered on a best-effort basis only.

9 research files, about 21,000 words



## PLAN

# Four milestones to December

**9 Oct**

### Pitch

Idea, use cases and mockups.

**11 Oct**

### 1st delivery

Design, research and fault model.

**13 Nov**

### Prototype

Clusters running, API, app and basic portal.

**18 Dec**

### Final delivery

Evaluated scheduler and fault demonstration.

### 104 tasks

GitHub Project, two-week sprints.

### Approved by the lecturers

The stack and the three nodes on one host.

### Division of work

Tiago: infrastructure, API, scheduler, security. César: portal, app, requirements, UML with Tiago, archive.





# Questions?

[github.com/Cabacadosinformaticos/fieldnote](https://github.com/Cabacadosinformaticos/fieldnote)

Tiago Cabaça · César Rodrigues  
IADE, Universidade Europeia · 2026-2027

